CSE/ISyE 6740
Computational Data Analysis

Midterm Review

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Kai Wang, Assistant Professor in Computational Science and Engineering
kwang692@gatech.edu

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Midterm Logistics

- Midterm is on **Oct 14th 12:30-1:45pm ET (75min) @ TSRB Room 132**
  - **Midterm exam scope**: everything until gradient descent (Lecture 12).

  - **Midterm format**:
    - Three question sets
    - Written (mostly math/proof/analysis/pseudo code)
    - No programming
    - Papers will be provided. Please bring your own pen/pencil with you.
    - Closed book, closed computer. **Calculator is not allowed nor needed.**
    - **One letter size/A4 hand-written note allowed (2-sided)**

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Midterm Review

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Overview

- **Unsupervised learning**
  - Clustering
  - Dimensionality reduction
  - Density estimation
  - EM algorithm

- **Supervised learning**
  - Feature selection and decision tree
  - Logistic regression
  - Support vector machines
  - Kernel methods
  - Neural networks and gradient descent

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Clustering

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Clustering Problem

- Given n data points: $\{x^1, x^2, \dots, x^n\} \in \mathbb{R}^d$
- Find k cluster centers: $\{c^1, c^2, \dots, c^k\} \in \mathbb{R}^d$
- Assign each data point to one cluster: $\pi(i) \in \{1,2, \dots, k\}$
- Such that the averaged square distances from each data point to its respective cluster center is small, i.e.,

$$\min_{c,\pi} \frac{1}{n} \sum_{i=1}^{n} \left\| x^i - c^{\pi(i)} \right\|^2$$

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K-means Algorithm

- Initialize k cluster centers $\{c^1, c^2, \dots, c^k\}$ randomly

- Do
  - Decide the cluster memberships of each data point, x^i, by assigning it to the nearest cluster center (cluster assignment)

$$\pi(i) = argmin_{j=1,2,\dots k} \left\| x^i - c^j \right\|^2$$

  - Adjust the cluster center (center adjustment)

$$c^j = \frac{1}{|\{i : \pi(i) = j\}|} \sum_{i:\pi(i)=j} x^i$$

- While(any cluster center has been changed)

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Dimensionality Reduction

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Principal Component Analysis

- Given n data points, $\{x^1, x^2, \dots, x^n\} \in \mathbb{R}^d$
- **Step 1**: estimate the mean and covariance matrix from the data:
  - $\mu = \frac{1}{n}\sum_{i=1}^{n} x^i$ and $C = \frac{1}{n}\sum_{i=1}^{n}(x^i - \mu)(x^i - \mu)^\top$
- **Step 2**: take the eigenvectors $w^1, w^2, \dots$of C corresponding to the largest eigenvalue λ_1, the second largest eigenvalue λ_2,…
- **Step 3**: compute reduced representation

$$z^i = \begin{pmatrix} \frac{w^{1\top}(x^i - \mu)}{\sqrt{\lambda_1}} \\ \frac{w^{2\top}(x^i - \mu)}{\sqrt{\lambda_2}} \\ \dots \end{pmatrix}$$

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Density Estimation and EM Algorithm

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Parametric v.s. Nonparametric

- Data $x \in \mathbb{R}^d$ with fixed dimension d
- Given n training data points $\{x^1, x^2, \ldots, x^n\}$
- Partition m bins in each dimension

Aspects	Gaussian and other distributions	Histogram	KDE
Flexible	No	Yes	Yes
Assumption	Strong	Mild	Mild
Parameter number	Fixed	Increased with m	Increased with n
Memory requirement	# parameters	m^d	nd
Training computation	Closed form	Binning and counting	Nothing
Test computation	Plug in formula	Find the bin	Evaluate n functions
Statistical guarantee	Only Gaussian (or other distributions)	Arbitrary (worse)	Arbitrary (better)

Details of EM

- We intend to learn the parameters that maximizes the log-likelihood of the data

$$l(\theta; D) = \log \prod_{i=1}^{n} \left(\sum_{Z^i=1}^{K} p(x^i, z^i|\theta) \right)$$

- **Expectation step (E-step)**: construct a good lower bound

$$l(\theta; D) \geq l(\theta, D; q) = \mathbb{E}_{z^1 \sim q^1, \ldots, z^n \sim q^n} \left[\log \prod_{i=1}^{n} p(x^i, z^i|\theta) \right] + H(q)$$

- **Maximization step (M-step)**: how to maximize?

$$\theta^{t+1} = argmax_\theta \, f(\theta)$$

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EM (Expectation-Maximization) Algorithm

- Associate each data and each component with a τ_k^i
- Initialized $(\pi_k, \mu_k, \Sigma_k), k = 1,2, \dots, K$
- Iterate the following two steps until convergence:
  - **Expectation step (E-step)**: update τ_k^i given the current (π_k, μ_k, Σ_k)

$$\tau_k^i = p(z^i = k \mid D, \mu, \Sigma) = \frac{\pi_k \, \mathcal{N}(x|\mu_k, \Sigma_k)}{\sum_{k'=1}^{K} \pi_{k'} \, \mathcal{N}(x|\mu_{k'}, \Sigma_{k'})}, \qquad \forall k \in \{1,2, \dots, K\}; i \in \{1,2, \dots, n\}$$

  - **Maximization step (M-step)**: update (π_k, μ_k, Σ_k) given τ_k^i

$$\pi_k = \frac{\sum_i \tau_k^i}{n}, \qquad \mu_k = \frac{\sum_i \tau_k^i x^i}{\sum_i \tau_k^i}$$

$$\Sigma_k = \frac{\sum_i \tau_k^i (x^i - \mu_k)(x^i - \mu_k)^\top}{\sum_i \tau_k^i}, \qquad \forall k \in \{1,2, \dots, K\}$$

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Feature Selection and Decision Tree

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Entropy: Quantify Uncertainty

- **Entropy** $H(Y)$ of a random variable Y

$$H(Y) = -\sum_{k=1}^{K} P(y=k)\log_2 P(y=k) \qquad \textbf{(Shannon entropy)}$$

$$H(Y) = \mathbb{E}_y[-\log_2 P(y)] = -\int_{y\in D} P(y)\log_2 P(y)\,dy \quad \text{(continuous case: } \textbf{differential entropy}\text{)}$$

- **Conditional entropy** $H(Y|X)$ of a random variable Y given X

$$H(Y|X) = -\int_{x\in D_X}\left(\sum_{k=1}^{K} P(y=k|x)\log_2 P(y=k|x)\right)p(x)dx$$

$$H(Y|X) = -\int_{x\in D_X}\left(\int_{y\in D_Y} P(y|x)\log_2 P(y|x)\,dy\right)p(x)dx \quad \text{(continuous case)}$$

- **Mutual information**: the reduction in uncertainty of Y after seeing feature X_i

$$I(X_i, Y) = H(Y) - H(Y|X_i)$$

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A Feature Selection Algorithm

- Given a dataset $D = \{(x^1, y^1), (x^2, y^2), \dots, (x^n, y^n)\}, x \in \mathbb{R}^d, y \in \{1,2, \dots, K\}$

- For reach value of the label $y = k$
  - Estimate density $p(y = k)$

- For the i-th feature entry x_i where $i \in \{1,2, \dots, d\}$
  - Estimate its density $p(x_i)$
  - For each value of the label $y = k$
    - Estimate the density $p(x_i|y = k)$
  - Score feature x_i using:

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_i \\ \dots \\ x_d \end{pmatrix} \in \mathbb{R}^d \text{ for each } x = x^j$$

$$I_i = I(Y; X_i) = \int \sum_{k=1}^{K} p(x_i|y = k)p(y = k) \log_2 \frac{p(x_i|y = k)}{p(x_i)} \, dx_i$$

- Choose those feature x_i with high score I_i

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Learning Decision Trees

- **Initialization**: start from empty decision tree with dataset D
- Based on D, compute information gain of each attribute (feature) X_i and split t

$$I(X_i > t, Y) = \arg\max_i H(Y) - H(Y|X_i > t)$$

- Pick **the best attribute (feature) and split t**

$$i, t := \arg\max_{i,t} I(X_i > t, Y) = \arg\max_{i,t} H(Y) - H(Y|X_i > t)$$

- Split the tree and the dataset at node $X_i > t$
- Create subtrees and recurse through all subtrees

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Logistic Regression

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Logistic Regression Objective $l(\theta)$

- Logistic regression model

$$p(y = 1|x, \theta) = \frac{1}{1 + \exp(-\theta^\top x)}$$

- Note that

$$p(y = 0|x, \theta) = 1 - \frac{1}{1 + \exp(-\theta^\top x)} = \frac{\exp(-\theta^\top x)}{1 + \exp(-\theta^\top x)}$$

- Plug in

$$l(\theta) \coloneqq \log \prod_{i=1}^{n} p\left(y^i \middle| x^i, \theta\right).$$

$$= \sum_i \left(y^i - 1\right)\theta^\top x^i - \log\left(1 + \exp\left(-\theta^\top x^i\right)\right)$$

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Support Vector Machines

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Soft-margin SVM Primal and Dual Problem

Primal problem:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi^i$$
$$s.t.\ 1-\xi^i - y^i\left(w^\top x^i + b\right) \le 0, \qquad -\xi^i \le 0 \quad \forall i$$

Dual problem:

$$\max_{\alpha} g(\alpha) := \sum_{i=1}^{n}\alpha_i - \frac{1}{2}\sum_{i,j=1}^{n}\alpha_i\alpha_j y^i y^j\left(x^{i\top}x^j\right)$$
$$s.t.\ C \ge \alpha_i \ge 0 \quad \forall i$$
$$\sum_{i=1}^{n}\alpha_i y^i = 0$$

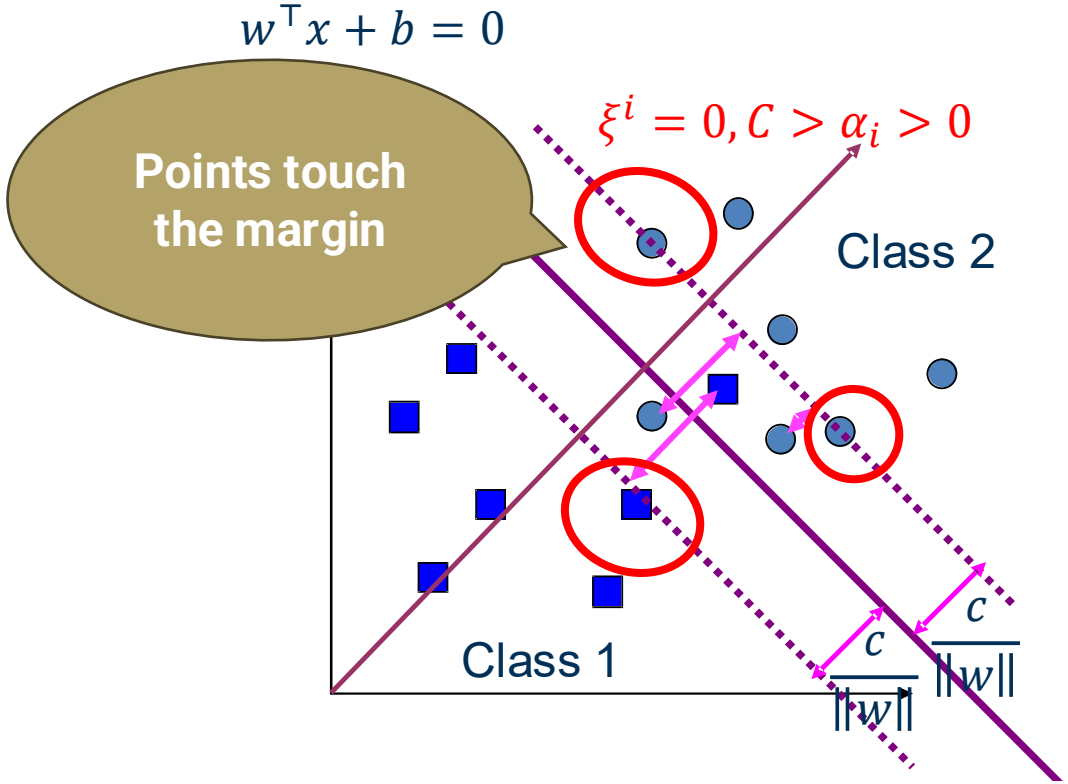
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Meaning of Dual Solution

- **Non-margin support vectors** $\alpha_i = C \neq 0$:
  - This means $\beta_i = C - \alpha_i = 0$
  - Therefore, $\xi_i > 0$, and $1 - y^i\left(w^\top x^i + b\right) = \xi^i > 0$

- **Margin support vectors** $0 < \alpha_i < C$:
  - This means $\beta_i = C - \alpha_i > 0$
  - Therefore, $\xi_i = 0$, and $1 - y^i\left(w^\top x^i + b\right) = \xi^i = 0$

- **Non-support vectors** $0 = \alpha_i$:
  - This means $\beta_i = C - \alpha_i = C > 0$
  - Therefore, $\xi_i = 0$, i.e., $1 - y^i\left(w^\top x^i + b\right) < \xi^i = 0$



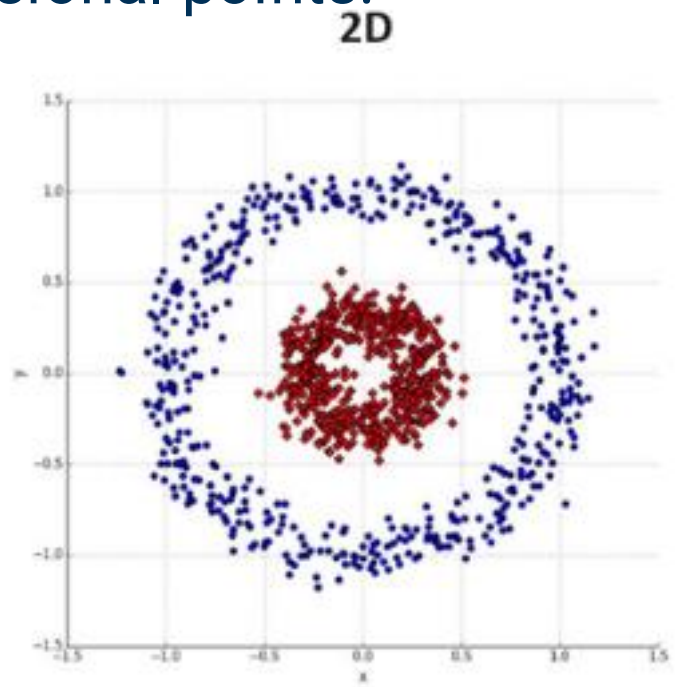
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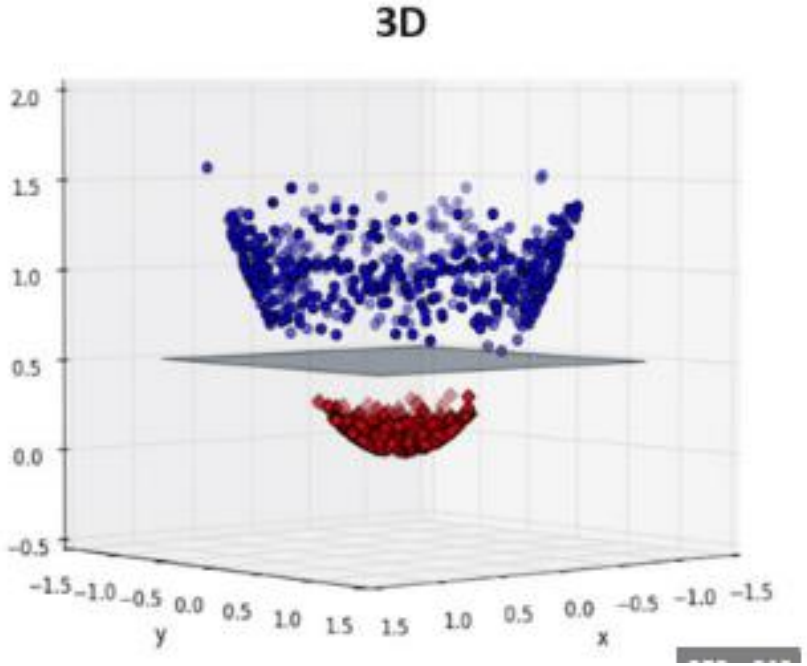
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Kernel Methods

- Define a non-linear mapping $\phi: \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$ to map from d-dimensional points to d'-dimensional points.





$x \in \mathbb{R}^d$ ⇨ $\tilde{x} = \phi(x) \in \mathbb{R}^{d'}$

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Avoid Using the Feature Mapping

- Use kernel methods

- Kernel $k: \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ measures the nonlinear similarity between two points $\mathrm{x}^{(i)}, x^{(j)}$
  - The main idea is to use $k\left(x^{(i)}, x^{(j)}\right) = \left\langle \phi\left(x^{(i)}\right), \phi\left(x^{(j)}\right) \right\rangle = \phi\left(x^{(i)}\right)^{\mathsf{T}} \phi\left(x^{(j)}\right)$ to replace all inner products.
  - Hope that we can just use the kernel function k to avoid ϕ

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Kernel Methods

- Kernel SVM

Dual problem + kernel k

$$\max_{\alpha} g(\alpha) := \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{n} \alpha_i \alpha_j y^i y^j k\left(x^{(i)}, x^{(j)}\right)$$

$$s.t. C \geq \alpha_i \geq 0 \quad \forall i, \qquad \sum_{i=1}^{n} \alpha_i y^i = 0$$

- Kernel regression

- **Ridge regression**: solve parameter $\theta^* = \left(\tilde{X}^\top \tilde{X} + \lambda I\right)^{-1} \tilde{X}^\top y$
  - $\Rightarrow$ prediction $\mathrm{x}^\top \theta^* = x^\top \left(\tilde{X}^\top \tilde{X} + \lambda I\right)^{-1} \tilde{X}^\top y$
- **Kernel regression**: prediction $k_x^\top (K + \lambda I)^{-1} y$

- Kernel PCA

PCA: $z^i = \begin{pmatrix} \frac{w^{1\top}(x^{(i)} - \mu)}{\sqrt{\lambda_1}} \\ \frac{w^{2\top}(x^{(i)} - \mu)}{\sqrt{\lambda_2}} \\ \dots \end{pmatrix}$ Kernel PCA: $z^i = \begin{pmatrix} \frac{w^{1\top}(\phi(x^{(i)}) - \mu)}{\sqrt{\lambda_1}} \\ \frac{w^{2\top}(\phi(x^{(i)}) - \mu)}{\sqrt{\lambda_2}} \\ \dots \end{pmatrix} = \begin{pmatrix} k_x^\top \alpha^1 \\ k_x^\top \alpha^2 \\ \dots \\ k_x^\top \alpha^n \end{pmatrix}$

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Gaussian Process Regression

- Bayesian linear regression

$$\mu_{pred} = x^{\top}(X^{\top}X + \sigma^2 S^{-1})^{-1}X^{\top}y$$
$$\sigma^2_{\text{pred}} = \sigma^2 x^{\top}(X^{\top}X + \sigma^2 S^{-1})^{-1}x + \sigma^2$$

- Gaussian process regression

$$\mu_{pred} = \phi(x)^{\top}\left(\tilde{X}^{\top}\tilde{X} + \sigma^2 I\right)^{-1}\tilde{X}^{\top}y$$
$$= k_x^{\top}(K + \sigma^2 I)^{-1}y$$

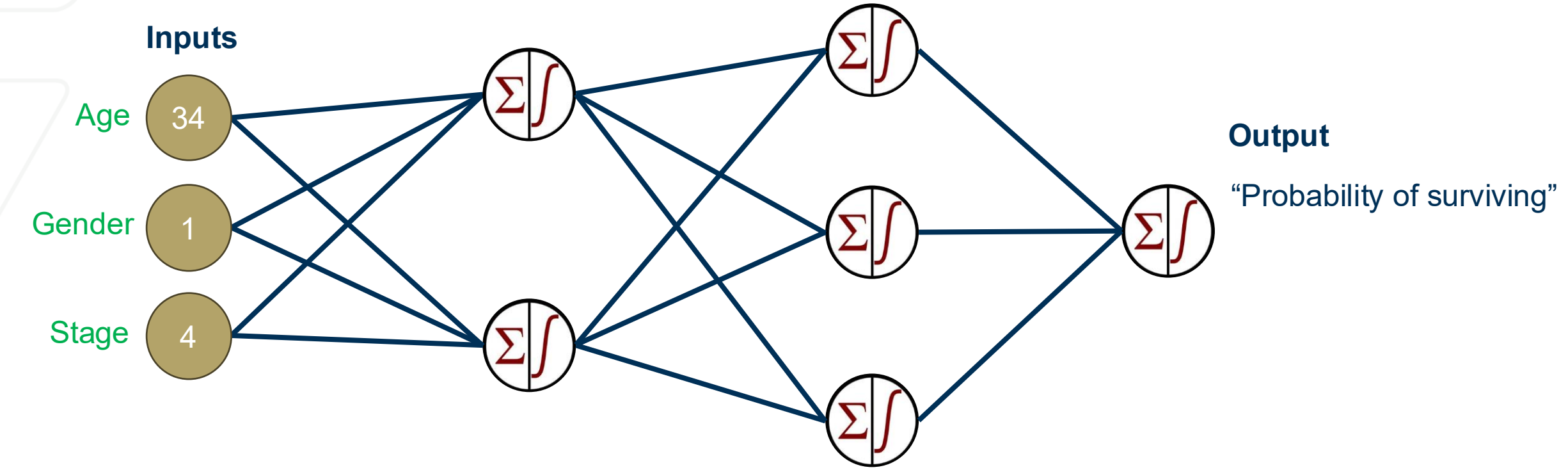
$$\sigma^2_{\text{pred}} = \sigma^2\phi(x)^{\top}\left(\tilde{X}^{\top}\tilde{X} + \sigma^2 I\right)^{-1}\phi(x) + \sigma^2$$
$$= k_{xx} - k_x^{\top}(K + \sigma^2)^{-1}k_x + \sigma^2$$

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Neural Network and Gradient Descent

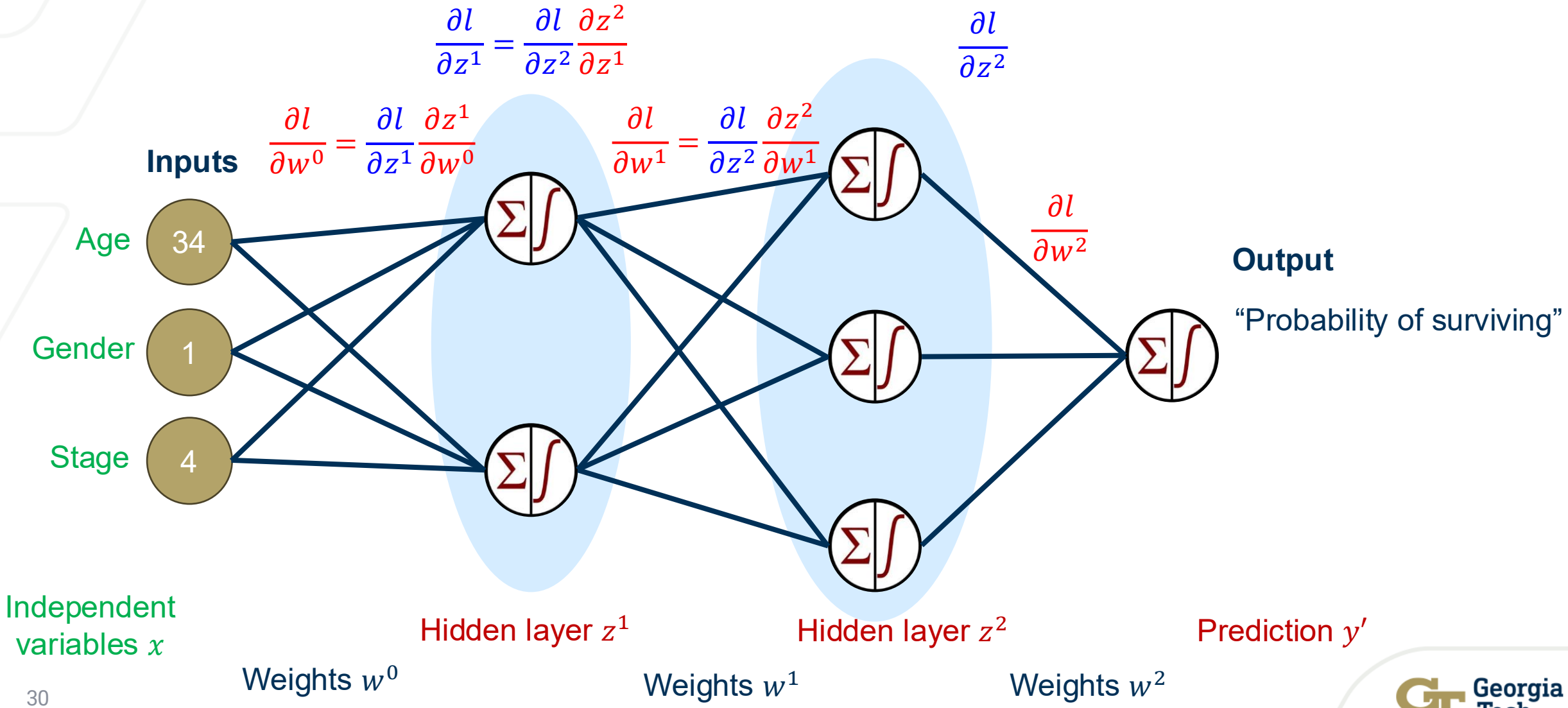
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Neural Network Model (Forward Pass)



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Compute the Gradient (Backpropagation)



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Gradient Descent

- First-order gradient descent

$$f(x+\delta) \approx f(x) + \nabla_x f(x)^\top \delta$$
$$x^{t+1} = x^t - \gamma \nabla_x f(x^t)$$

- Second-order gradient descent

$$f(x+\delta) \approx f(x) + \nabla_x f(x)^\top \delta + \frac{1}{2}\delta^\top \nabla_x^2 f(x)\delta + o(\|\delta\|^2)$$
$$x^{t+1} = x^t - \left(\nabla_x^2 f(x^t)\right)^{-1} \nabla_x f(x^t)$$

- Adaptive gradient descent

$$x^{t+1} = x^t - \gamma (G^t)^{-\frac{1}{2}} \nabla_x f(x^t)$$

- Momentum-based method

$$z^t = \beta z^{t-1} + \nabla_x f(x^t)$$
$$x^{t+1} = x^t - \gamma z^t$$

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- Bring your favorite pens and pencils.
- See you on Wednesday!

